

Ish Thumber

ishthumber343@gmail.com | +91 9909919803 | ishthumber.vercel.app | linkedin.com/in/ishthumber | github.com/IshThumber
medium.com/@ishthumber

SUMMARY

Cloud Engineer (Python, FastAPI, React) specialising in AI-driven systems. Delivers significant efficiency gains by accelerating development cycles and optimising cloud environment setup.

PROFESSIONAL EXPERIENCE

Cloud Engineer
Searce Inc.

March 2024 – Present | Hyderabad, India

- Designed reusable, modular Terraform IaC components that standardised environment provisioning across multiple client projects, reducing infrastructure duplication by 30% and accelerating team onboarding efficiency.
- Migrated large-scale GCP Apigee infrastructure using Terraform, implementing Private Service Connect (PSC) architecture to replace legacy systems and enable seamless multi-environment scalability.
- Engineered a cross-region disaster recovery strategy (Mumbai to Hyderabad) using AWS Lambda and EC2 Data Lifecycle Manager with RDS Point-In-Time Recovery, achieving a 15-minute RTO for critical workloads.
- Implemented an observability stack using Prometheus, Grafana, Loki, and Alertmanager to monitor 150+ EC2 instances with custom alert rules, reducing MTTR by 40%.
- Designed high availability Fortinet firewall architectures on AWS with automated failover mechanisms, achieving 99.99% uptime for enterprise security workloads.
- Deployed production-grade Kubernetes monitoring on Amazon EKS with Prometheus, Grafana, and Alertmanager for automated incident notifications while implementing RBAC policies to strengthen cluster security.

SKILLS

- **Programming Languages:** Python, JavaScript, C++
- **Cloud Platforms:** Amazon Web Services (EC2, S3, IAM, VPC), Microsoft Azure (Azure Blob Storage, Azure Virtual Machines), Google Cloud Platform (Google Cloud Storage, Compute Engine)
- **IaC & Automation:** Terraform, Helm, Jenkins, GitHub Actions, AWS CodePipeline, AWS CodeCommit, AWS CodeBuild
- **Monitoring & Logging:** Prometheus, Grafana, Loki, AlertManager, CloudWatch
- **Containerisation & Orchestration:** Docker, Kubernetes, Amazon Elastic Kubernetes Service
- **Web Development:** React.js, Node.js, Express.js, Tailwind CSS, HTML, CSS
- **Core Subjects:** Data Structures and Algorithms, Cloud Computing, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Object-Oriented Programming

PROJECTS

Observability Accelerator

September 2025 – Present

- Reduced monitoring setup time from hours to under 10 minutes by architecting a React 19 guided deployment wizard with Server Actions and use hooks, eliminating manual configuration errors across 20+ microservices and improving developer productivity organisation-wide.
- Implemented real-time deployment visibility via Server-Sent Events (SSE), streaming 20+ pipeline stage updates to a terminal-style UI with execution timing, stage-level status, and actionable failure diagnostics for complex async workflows.
- Architected an AWS metrics integration layer featuring CloudWatch metrics selection with tag-based filtering and automatic service discovery (EC2, RDS), enabling seamless ingestion of AWS-native telemetry into Prometheus without manual exporter configuration.
- Delivered unified observability dashboards combining metrics (Prometheus), logs (Loki), traces (Tempo), and CloudWatch with preconfigured alerts and multi-channel notifications to accelerate incident response and reduce MTTR.
- Orchestrated end-to-end full-stack workflows handling async state, retries, and partial failure recovery across backend services, while implementing optimistic UI patterns to maintain 100% state accuracy during long-running infra provisioning.
- Established CI/CD pipelines and end-to-end testing frameworks to automate deployment validation, ensuring reliable delivery of the complete observability stack from development to production.

RAG Flashcards Generator

August 2025 – September 2025

- Developed a scalable, serverless backend using FastAPI with asynchronous I/O and modular services, deployed on AWS Lambda and API Gateway.
- Engineered a complete Retrieval-Augmented Generation (RAG) pipeline, achieving sub-second semantic search by integrating AWS Textract, Amazon Titan embeddings (1024-dim), and OpenSearch kNN.
- Incorporated Claude 3.5 Sonnet for generating context-aware flashcards, ensuring high factual accuracy through citation tracking and retrieval grounding.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03)
Terraform Certified Associate (HCTAO-003)
AWS Certified Cloud Practitioner (CLF-02)

RECOGNITION

AWS Community Builder – DevTools
AWS

Recognized by AWS for contributions to cloud automation, DevOps, and infrastructure development using tools like CodePipeline, Terraform, and Kubernetes.

EDUCATION

Bachelors in Technology, Computer Engineering
Charotar University of Science and Technology
CGPA: 8.5

July 2020 – May 2024